25901442 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY203 - OPTOELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What is meant by reverse saturation current? [K] / [CO1]
2. Does the mass action law hold true for degenerate semiconductors? Why? [U] / [CO1]
3. How is a p-type semiconductor formed? Explain with an example. [U] / [CO1]
4. What is a Phonon? [U] / [CO1]
5. State the expression for the effective density of states in the conduction band edge? [U] / [CO1]
6. List any two semiconductor materials used to fabricate LEDs. [K] / [CO2]
7. Why are LEDs suitable for optical fiber communication? [U] / [CO2]
8. What is the basic principle of light emission in an LED? [K] / [CO2]
9. What is the role of phosphor in white LEDs? [U] / [CO3]
10. What is meant by noise equivalent power (NEP). [K] / [CO3]

11. What is the basic working principle of a solar cell? [K] / [CO3]

12. What is the purpose of an antireflection coating on a photodiode? [U] / [CO3]

13. Define signal-to-noise ratio (SNR). [K] / [CO3]

14. What change occurs in the I-V characteristics of a photodiode when illuminated? [U] / [CO3]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Describe diffusion capacitance and its importance in forward bias operation. [U] / [CO1]

16. Explain the Shockley diode equation and the significance of each term. [U] / [CO1]

17. Explain the formation of a pn junction under open-circuit (equilibrium) condition. [U] / [CO1]

18. Compare the internal and external quantum efficiency of an LED [K] / [CO2]

19. Explain the structure and operation of a mutiple quantum well LED. [U] / [CO2]

20. Explain the absorption in direct and indirect band gap materials [U] / [CO3]

21. Explain the processes of carrier generation, drift, and collection in the intrinsic region of a PIN photodiode. [K] / [CO3]

22. Write a note on multijunction tandem solar cell. [U] / [CO3]

[5x6 = 30]